

LabExercise 05**IT3060– Human Computer Interaction****Semester 2 2026**

Please refer to the given case study on CourseWeb and complete the lab sheet accordingly.

Activity 1:**Compare Flutter, React Native, Kotlin and Swift**

- Analyze the strengths and weaknesses of Flutter, React Native, Kotlin Multiplatform, and Swift/SwiftUI for mobile and cross-platform development.
- Compare them across key criteria: development speed, code reusability, performance, ecosystem support, learning curve, web compatibility, AI/ML integration, real-time features, maintenance cost, and security.
- Evaluate suitability for a fitness app requiring seamless iOS/Android/web experience and high performance.
- Prepare a recommendation with detailed justification for the best option (or hybrid approach) for the FitFlow redesign.

Activity 2:**Compare Backend, Database and Authentication Options**

- Research and compare popular backend frameworks (Node.js/NestJS, Python/FastAPI, Go, etc.).
- Compare database options (PostgreSQL, MongoDB, Firebase, DynamoDB, etc.) focusing on scalability, query performance, and health data handling.
- Evaluate authentication and authorization solutions (Firebase Auth, AWS Cognito, Auth0, Supabase, etc.).
- Assess each option for security (HIPAA/GDPR compliance), real-time capabilities, AI integration, cost, and maintainability for a mid-sized team.
- Recommend the most suitable combination with clear justifications.

Activity 3:**Prepare a Technology Comparison Matrix**

- Consolidate findings from frontend, backend, database, and authentication comparisons.
- Create a comprehensive decision matrix with scoring across relevant criteria (performance, scalability, development speed, security, cost, AI/ML support, maintainability, etc.).
- Apply appropriate weights to each criterion based on FitFlow project needs.
- Highlight the recommended technology stack and provide supporting rationale.

Activity 4:**Design a High-Level Architecture**

- Design the overall system architecture using the selected technology stack.
- Define key components including frontend, backend services, AI microservice, databases, caching, and real-time layers.
- Illustrate data flows for critical features such as personalized workout plans, social sharing, and nutrition tracking.
- Document security, scalability, and integration considerations.
- Create a visual diagram and a short Architecture Decision Record (ADR).

Activity 5:**Create a GitHub Repository**

- Create a new GitHub repository named fitflow-redesign.
- Set up the initial project folder structure (frontend, backend, ai-service, docs, etc.).
- Add essential documentation including README.md, tech stack summary, comparison matrix, and architecture diagram.
- Configure basic repository settings (branch protection, .gitignore, optional CI/CD workflow).
- Make the first commit with all supporting documents from previous activities.

Submission Note: Complete each activity separately. Include frontend/backend/database/authentication comparison tables, the weighted decision matrix, recommended technology stack justification, high-level architecture diagram, ADR, GitHub repository link, README, and all supporting documents in your lab report.